

Joseph Lee

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EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, Canada

Sept. 2024 – June 2029

- President's Scholarship of Distinction
- Relevant Coursework: OOP, Data Structures & Algorithms, Combinatorics, Optimization

EXPERIENCE

Software QA Analyst

Sept. 2025 - Dec. 2025

Environment and Climate Change Canada

Toronto, Canada

- Engineered automation frameworks in Python, Robot Framework, Selenium, and RPAFramework across 100+ national applications supporting Canada's Vision 2050 climate goals, cutting manual test cycles by 60%
- Built Azure SQL validation pipelines verifying backend data, REST APIs, and large datasets feeding analytics and ML workflows, catching schema drift and data integrity issues pre-production
- Shipped Azure DevOps dashboards tracking defects, performance metrics, and regression coverage, enabling data-driven release decisions and driving defect triage that lowered duplicate report rate by 40% across the team
- Executed functional and cross-browser test campaigns for the National Air Pollution Surveillance (NAPS) program, surfacing 20+ defects pre-release and validating system scalability and reliability across cloud-hosted deployments

IT Technician and Support Analyst

May 2025 - Dec. 2025

Environment and Climate Change Canada

Toronto, Canada

- Delivered Tier 1/2 support for 200+ end users across Windows, Linux, M365, and Outlook server environments, cutting average resolution time by 50% through reusable diagnostic playbooks and root cause analysis
- Administered Active Directory, Microsoft Admin Center, and Entra ID, managing authentication, RBAC permissions, group policy, and compliance for users across multiple departments and contractor pools
- Engineered Python, PowerShell, Power Automate, and SAP GUI scripting automations to consolidate repetitive cross-system admin tasks, shortening end-to-end task duration by 80% across daily IT operations
- Deployed and hardened Windows workstations at scale using Autopilot, Intune, and Entra ID, standardizing setup across the team and shrinking per-device manual configuration time from hours to minutes

ServiceNow Software Developer

Jan. 2025 - Aug. 2025

ServiceNow Developer Program

Toronto, Canada

- Shipped Client Scripts and UI Policies automating multi-step approval workflows on the ServiceNow platform, reducing manual steps and tightening process consistency across business operations
- Leveraged GlideForm and GlideUser APIs for dynamic validation and role-based UI behavior, debugging client-side JavaScript across forms and workflows to ensure reliable behavior, API compliance, and consistent data quality

PROJECTS

Image Lab | *Python, FastAPI, PyTorch, OpenCV, Docker, CustomTkinter* | [GitHub](#)

2025

- Architected a multi-interface image processing platform (FastAPI backend, web frontend, desktop GUI) sharing a single processing core, supporting 13+ filters and ML-powered transformations
- Integrated U2-Net for background removal and transfer-learned ResNet/EfficientNet for auto-tagging; containerized with Docker and exposed a documented REST API with Pydantic schemas
- Engineered a pytest-covered plugin-pattern filter pipeline for seamless integration of new transformations

Chess Engine | *C++, MVC, Observer, Decorator, chrono* | [GitHub](#)

2025

- Built a fully compliant C++ chess engine with complete rule enforcement including castling, en passant, promotion, and check/checkmate detection
- Applied MVC, Observer, and Decorator patterns to cleanly decouple game logic, rendering, and state management
- Implemented minimax AI with difficulty levels and `std::chrono` per-player timers, achieving 105% course grade

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript/TypeScript, SQL, R, PowerShell

Frameworks: React, Node.js, FastAPI, Flask, PyAutoGUI, Playwright, Robot Framework

Libraries: Pandas, NumPy, SciPy, Matplotlib, TensorFlow, PyTorch, Pygame

Developer Tools: Git/Jira, Docker, Azure DevOps, GitLab, ServiceNow, SAP S/4 HANA, Power Automate